

PARAPARAUMU, New Zealand

WMO: 934200

Lat: 40.9040S

Lon: 174.9840E

Elev: 8

StdP: 101.23

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.3	1.6	-1.7	3.3	2.7	-0.6	3.6	3.4	12.0	12.1	10.9	11.6	1.5	100	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.0	23.2	18.2	22.1	17.7	21.2	17.3	19.4	21.7	18.7	20.9	18.2	20.4	4.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	20.5	17.9	12.8	20.1	17.1	12.3	19.7	55.4	21.9	53.3	21.0	51.3	20.4	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.8	-1.9	25.8	0.7	1.7	-2.4	27.1	-2.8	28.1	-3.2	29.1	-3.7	30.3		
(5)			-2.4	20.6	0.7	0.9	-2.9	21.3	-3.3	21.8	-3.7	22.4	-4.3	23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	17.1	17.3	15.9	13.6	12.0	9.7	8.9	9.4	10.9	12.2	13.6	15.9
	DBStd	3.74	2.16	2.26	2.40	2.60	2.73	2.79	2.58	2.32	2.35	2.15	2.11	2.16
	HDD10.0	176	0	0	0	2	12	41	54	40	18	7	1	0
	HDD18.3	1973	49	42	81	141	197	259	294	276	222	190	142	80
	CDD10.0	1278	221	203	182	112	74	32	19	22	47	76	109	182
	CDD18.3	33	11	13	4	1	0	0	0	0	0	0	0	4
	CDH23.3	28	11	9	4	0	0	0	0	0	0	0	0	5
	CDH26.7	1	0	1	0	0	0	0	0	0	0	0	0	1
Wind	WSAvg	4.3	4.5	4.2	4.1	3.7	4.1	3.9	3.9	4.2	4.6	4.9	5.0	4.6
Precipitation	PrecAvg	1114	72	73	73	79	96	114	106	101	92	122	90	96
	PrecMax	1534	132	388	210	181	196	173	246	171	230	356	184	170
	PrecMin	798	19	18	22	30	21	39	44	24	22	49	10	38
	PrecStd	162	31	73	38	41	45	39	48	40	44	57	50	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	24.5	23.6	21.0	18.8	16.5	15.3	15.4	16.8	18.6	21.3	23.6
		MCWB	19.3	18.5	18.1	16.9	15.2	13.8	12.4	12.1	12.9	14.4	16.5	17.9
	2%	DB	23.1	23.0	21.9	19.5	17.6	15.5	14.5	14.5	15.7	17.0	19.2	21.5
		MCWB	18.4	18.4	17.5	16.3	15.2	13.3	11.9	11.8	12.7	13.9	15.3	17.1
	5%	DB	21.8	22.0	20.8	18.6	16.8	14.8	13.8	13.8	15.1	16.2	18.0	20.4
		MCWB	17.7	17.9	16.9	15.6	14.5	12.6	11.4	11.4	12.5	13.4	14.6	16.7
	10%	DB	20.9	21.1	20.0	17.8	16.1	14.1	13.1	13.2	14.5	15.5	17.0	19.4
		MCWB	17.2	17.5	16.6	15.1	13.9	12.0	10.8	10.9	12.1	12.9	13.9	16.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	20.6	19.4	18.1	16.8	14.7	13.6	13.4	14.4	15.6	17.2	19.2
		MCDB	23.2	22.2	22.0	19.8	17.9	15.7	14.7	14.7	15.6	17.1	19.8	21.8
	2%	WB	19.3	19.6	18.5	17.1	15.7	13.8	12.6	12.5	13.7	14.6	16.2	18.2
		MCDB	21.7	21.5	20.8	18.6	17.0	15.0	13.9	13.8	14.9	16.2	18.3	20.2
	5%	WB	18.6	18.9	17.8	16.4	15.0	13.0	11.8	11.8	13.1	14.0	15.4	17.5
		MCDB	20.8	21.0	20.0	18.1	16.6	14.5	13.4	13.4	14.5	15.7	17.2	19.5
	10%	WB	17.9	18.2	17.1	15.6	14.2	12.3	11.1	11.2	12.5	13.4	14.6	16.8
		MCDB	20.2	20.5	19.5	17.5	15.9	13.9	12.8	12.9	14.1	15.2	16.4	18.9
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.0	7.3	7.2	6.5	6.6	6.9	6.6	6.1	5.8	6.0	6.0
		MCDBR	7.8	7.9	8.3	7.1	6.0	5.6	6.3	6.6	6.1	6.1	7.0	7.1
		MCWBR	4.6	4.8	5.2	5.0	4.6	4.6	4.9	5.0	4.5	4.4	4.7	4.2
	5% WB	MCDBR	6.2	6.0	6.4	5.4	5.0	4.7	5.2	5.4	4.8	5.0	5.5	5.5
		MCWBR	4.2	4.1	4.6	4.4	4.2	4.2	4.4	4.5	4.1	3.9	4.2	3.7
Clear-Sky Solar Irradiance	taub	0.350	0.337	0.323	0.312	0.300	0.297	0.295	0.305	0.328	0.342	0.334	0.345	
	taud	2.533	2.590	2.627	2.644	2.645	2.624	2.629	2.613	2.527	2.502	2.546	2.550	
	Ebn at Noon	979	966	939	884	829	795	825	878	916	950	988	989	
	Edh at Noon	108	97	86	73	62	58	62	74	93	105	106	107	
All-Sky Solar Radiation	RadAvg	6.72	5.91	4.63	3.09	2.07	1.55	1.81	2.62	3.79	5.06	6.28	6.55	
	RadStd	0.43	0.51	0.27	0.23	0.11	0.09	0.13	0.15	0.24	0.35	0.49	0.45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)		N/A	N/A	+0.41	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page